

Thenmozhi Boopathy

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PROFESSIONAL SUMMARY

Career-changer from a math-teaching background applying that analytical rigor to data work — built end-to-end ML pipelines spanning classification, forecasting, deep learning, and agentic RAG using Python, R, and Apache Spark.

EDUCATION

Miami University

Master of Science in Business Analytics

U.S. Permanent Resident (Green Card) - no visa sponsorship required

Oxford, OH

Expected Graduation 2027

Bharathidasan University

Master of Philosophy in Mathematics

Master of Science in Mathematics

Bachelor of Science in Mathematics

TamilNadu, India

2008

2007

2005

TECHNICAL SKILLS

Certification: Databricks University Alliance Student — Databricks (issued 2026). TNSET (Tamil Nadu State Eligibility Test) — Assistant Professor Eligibility, Mathematical Sciences (2017).

Languages & Tools: Python, R, SQL, Jupyter Notebook, RStudio, Visual Studio Code, GIT

Data Platforms: Databricks, Apache Spark, Spark MLlib, Delta Lake, NoSQL (MongoDB), Power BI

ML / AI: Pandas, Data & ML Pipelines, LangChain, MLflow, Vector Search, GenAI / LLM, Agentic AI, Deep Learning, NLTK

RELEVANT COURSEWORK

Data Mining: Exploratory Data Analysis, Dimension Reduction, Tree-Based Models, Logistic Regression, Neural Networks, Model Evaluation,

Big Data: Core Python, Pandas, NoSQL (MongoDB), Web APIs, Scikit-learn, Spark MLlib, Text Mining (NLTK, TF-IDF, Word2Vec).

Cloud & GenAI: Databricks, Apache Spark, Delta Lake, Unity Catalog, Databricks Vector Search, LangChain, MLflow, Agentic AI, RAG.

Current Coursework (Fall 2026): Power BI & DAX, Data Storytelling & Dashboards, Linear/Integer Programming (Gurobi), Reproducible Analytics & Business Value Proposition.

PROJECT EXPERIENCE

Enterprise RAG Chatbot — Job Finder Assistant (ISA-632)

2026

Built and deployed an enterprise-grade Retrieval-Augmented Generation (RAG) chatbot on Databricks with a 3-person team: used Apache Spark to ingest 2,500+ job records into Delta Tables, integrated Databricks Vector Search with a LangChain agent on Llama 4 Maverick, and built a Streamlit UI with MLflow-tracked evaluation, achieving 89% positive user feedback.

End-to-End R Machine Learning Workflow for Loan Default Prediction (ISA-591)

2025

Built an end-to-end R pipeline (decision trees, logistic regression, neural network, XGBoost) to predict loan default on an imbalanced dataset (80.8% no-default / 19.2% default); diagnosed class imbalance behind a 0.069 baseline sensitivity, then raised it to 0.64 (AUC 0.74) via downsampling and cross-validated pruning, and tuned the decision threshold with Youden's J to minimize costly false negatives.

Theme Park Attendance Forecasting Pipeline with Python (ISA-514)

2025

Built a Python-based machine learning pipeline using MongoDB for data storage and integration of structured and unstructured sources (weather, wait times, schedules, Reddit sentiment) to forecast Six Flags attendance for operational decision-making.

PROFESSIONAL EXPERIENCE

S.A College of Engineering

Assistant Professor

Chennai, India

Jan–May 2015

- Delivered instruction in quantitative and statistical methods (numerical analysis, probability, matrices, and applied functions), emphasizing analytical problem-solving skills relevant to data-driven business decision-making.

Sri Venkateswara College of Engineering

Assistant Professor

Chennai, India

2012-2013

- Performed lecturing in advanced quantitative methods (calculus, sequences, integrals), building strong analytical and problem-solving foundations relevant to data modeling and forecasting.

Bon Secours College for Women

Lecturer (PG and UG)

Tanjore, India

2009-2012

- Conducted training in mathematics and analytics concepts for undergraduate and postgraduate students, strengthening quantitative reasoning, data interpretation, and problem-solving skills.

VOLUNTEER ACTIVITIES

Language Teacher

Greater Cincinnati Tamil School, Cincinnati

Cincinnati, Ohio

2021-2025

- Designed and provided customized language programs for students of varying proficiency levels, using assessments to track progress and enhance outcomes. Coordinated program logistics and engaged stakeholders to promote the Seal of Biliteracy.

Gifted Education Co-Representative

Sycamore Gear

Cincinnati, Ohio

2023-2024

- Designed and executed student engagement initiatives, enhancing community involvement and program effectiveness.